Niural AI

AI ENGINEER CASE STUDY

Congratulations, you are a candidate for joining the Niural team!

Background:

Build a customer support agent for Niural that answers product and account questions through conversation. Choose your own architecture and framework. Choose from the runtime models available through the supplied OpenAI access.

Your Task:

- Answer questions from the supplied documentation, with source references.
- Handle follow-ups and corrections, and use the mock account API when needed.
- Keep customer conversations isolated; handle ambiguity, missing information, and tool errors through clarification or appropriate escalation.

Resources:

We will provide help articles, a mock API with synthetic account data, sample conversations, and an evaluation interface. Use these materials as the source of truth; no production access is needed.

Niural will provide a temporary OpenAI API key with a $10 usage allowance. We will evaluate your submitted model configuration. Credentials and setup instructions will be shared separately.

Deliverables:

- Runnable code and setup instructions, compatible with the supplied evaluation interface, including source references and observable tool calls. Python preferred; a CLI is enough.
- **Evaluate and improve your agent:** Create a small test set covering answer quality, multi-turn corrections, and tool failures. Record initial results, inspect a failure using execution traces, and test a targeted change. Include initial and revised code versions, before-and-after results using the same test cases and scoring criteria, and relevant execution traces. Explain regressions and whether you would keep the change; a justified rollback or inconclusive result is acceptable. AI-assisted or automated iteration is welcome.

- **Live presentation and demonstration:** Walk us through your approach, assumptions, tradeoffs, evaluation results, and limitations. Have your agent ready to run: interviewers will ask it unseen questions and multi-turn follow-ups within the supplied documentation and mock API's scope. Be prepared to explain its behavior, investigate failures, and demonstrate a small change.

Evaluation:

Niural has its own private benchmarking dataset, which we will use to evaluate your agent. It will not be shared with candidates.

The benchmark tests the requirements above using unseen questions and conversations. We will also assess engineering quality, validation, latency, model usage, and your ability to explain and improve the solution.

Time & AI Tools:

You have a full day to complete the case study. Estimated completion time: < 4 hours, excluding setup. Partial completion is acceptable; explain unfinished work. You may use your preferred AI coding tools—describe how you used them and checked their output. No frontend or deployment required.